

Multi-Agent Analytics Report

June 01, 2026

Executive Summary

- Decision: Approve and fund the 60-day overtime audit in Sales and the 90-day travel policy review as the highest-priority, evidence-based actions. These targeted initiatives address the most extreme attrition drivers identified in the data. Simultaneously, empower HR and Sales VPs to conduct local diagnostics. This combined approach addresses systemic issues (travel/overtime) and local context, providing a rapid and responsible path to reducing turnover costs and stabilizing key teams.
- Who affected: records or segments tied to the strongest observed Attrition risk signals.
- Evidence: Sales department attrition is 25.8%, highest among departments.
- Pilot metric: track outcome lift, precision, recall, F1/support, and implementation quality where screening is used.
- Caveat: The predictive model's accuracy (81.9% test accuracy) is insufficient for reliably scoring individual employee attrition risk. Using it for individual decisions could lead to false positives/negatives and fairness issues. Mitigation: Use the model's rules (e.g., Overtime='Yes', Travel_Frequently='Yes') only to define high-risk *groups* for program targeting. Do not use it for performance management or individual interventions. Validate any group-level action with direct manager feedback.

User context: hr attrition

Analysis of employee attrition data reveals that while the company's overall attrition rate (16.1%) is favorable compared to the UK market average (19%), severe, concentrated risks exist within specific employee segments. Overtime and frequent travel are the strongest predictors of turnover, with attrition rates exceeding 24%. The Sales and Human Resources departments are particularly vulnerable. Our interpretable model identifies these high-risk patterns but is not yet accurate enough for individual employee predictions. Therefore, we recommend immediate, targeted interventions focused on workload management, travel policy review, and support for high-risk departments, rather than broad policy changes or individual risk scoring.

Unplanned employee attrition drives significant direct costs (recruitment, onboarding) and indirect costs (lost productivity, institutional knowledge). The business objective is to stabilize the workforce by proactively retaining talent. This analysis moves beyond the aggregate 'attrition rate' KPI to identify the 'who' and 'why' behind turnover, enabling efficient allocation of retention resources to the segments and drivers with the highest proven impact.

Objective Coverage

- Objective: hr attrition
- Analysis alignment: Analysis identifies key drivers of employee attrition through EDA and interpretable decision tree rules, providing actionable insights for retention strategies.
- Decision tree target: Attrition
- Decision answer: Approve and fund the 60-day overtime audit in Sales and the 90-day travel policy review as the highest-priority, evidence-based actions. These targeted initiatives address the most extreme attrition drivers identified in the data. Simultaneously, empower HR and Sales VPs to conduct local diagnostics. This combined approach addresses systemic issues (travel/overtime) and local context, providing a rapid and responsible path to reducing turnover costs and stabilizing key teams.

Dataset Overview / Data Understanding

The HR attrition dataset is high-quality with complete data, supporting detailed analysis of employee turnover. It enables identification of key attrition drivers through exploratory analysis and interpretable decision tree modeling. Recommendations should focus on actionable retention strategies, noting that outliers in income and tenure may require careful handling. Constant variables should be excluded to ensure model effectiveness.

WA_Fn-UseC_-HR-Employee-Attrition.csv: Dataset has 1470 rows and 35 columns with no missing values or duplicate rows. Data is clean but has outliers in several numeric columns such as MonthlyIncome (7.76% outliers), TrainingTimesLastYear (16.19% outliers), and YearsAtCompany (7.07% outliers). Recommended analyses: Exploratory Data Analysis (EDA) to visualize distributions and relationships between features and Attrition rates., Decision tree classification model to predict Attrition and extract interpretable rules for business insights., Correlation analysis to identify strong associations between numeric variables and Attrition..

Market Research

The provided dataset contains HR employee attrition data, which is a critical business issue across industries. Current market data shows attrition rates vary significantly by region and sector, with professional services (consulting, legal, IT) experiencing particularly high turnover. The market is shifting from reactive to predictive management approaches, with AI and machine learning becoming key tools for early risk identification.

- The average attrition rate in the UK in 2025 was 19%, above the European average of 17.4%, representing an 11% increase from 2024. [0]

Source [0]: Retention trends 2026: attrition rates data and employee retention strategies - <https://ravio.com/blog/employee-retention-trends>

- For India's top-5 IT services companies, the attrition rate has stabilized at approximately 13% as of FY2025, significantly lower than pre-pandemic levels of 18%. [1]

Source [1]: Attrition Rate in India 2026: Trends & Industry Data - <https://www.wisemonk.io/blogs/managing-attrition-in-india>

- Consulting, legal, and IT roles have traditionally experienced high turnover, with professional services companies seeing attrition outpace hiring. [2]

Source [2]: Employee turnover by industry: Cost of attrition | Achievers - <https://www.achievers.com/blog/employee-turnover-by-industry/>

- Artificial intelligence is transforming attrition management from reactive to predictive, with machine learning algorithms analyzing communication patterns, performance metrics, and behavioral data. [4]

Source [4]: Attrition Management [2026]: Strategies, Best Practices, Examples - <https://taggd.in/blogs/attrition-management/>

Analysis Plan

- Identify key factors driving employee attrition to inform targeted retention strategies
- Develop interpretable decision rules that predict attrition risk for business decision-makers
- Quantify the relationship between employee attributes, work conditions, and attrition likelihood
- Descriptive statistics for attrition rate comparison against market benchmarks (UK: 19%, India IT: 13%)
- Chi-square tests for categorical predictor associations with attrition
- T-tests or Mann-Whitney U tests for numeric variable differences between attrition groups
- Decision tree classification with Gini/entropy criterion, limited depth for interpretability
- Model evaluation metrics: accuracy, precision, recall, F1-score, ROC-AUC with focus on minority class (attrition)

Data Quality Notes

- WA_Fn-UseC_-HR-Employee-Attrition.csv: Remove or flag constant columns: EmployeeCount, Over18, and StandardHours have only one unique value and provide no predictive power for analysis.
- WA_Fn-UseC_-HR-Employee-Attrition.csv: Consider outlier treatment for columns like MonthlyIncome, TrainingTimesLastYear, and YearsAtCompany if they skew statistical models or visualizations.
- WA_Fn-UseC_-HR-Employee-Attrition.csv: EmployeeNumber is a unique identifier and should be excluded from predictive models to avoid overfitting.

- WA_Fn-UseC_-HR-Employee-Attrition.csv: Attrition is a binary target variable (Yes/No) suitable for classification analysis.
- WA_Fn-UseC_-HR-Employee-Attrition.csv: MonthlyIncome has 7.76% outliers based on IQR method, which may indicate high earners or data errors.
- WA_Fn-UseC_-HR-Employee-Attrition.csv: TrainingTimesLastYear has 16.19% outliers, suggesting variability in training participation that could affect attrition.

Data Analysis and Visual Findings

- Overall attrition rate is 16.1%, higher than typical UK average (19%) and India IT sector (13%).
- Sales department shows highest attrition (20.6%), suggesting role-specific stressors.
- Employees traveling frequently have 24.9% attrition rate, indicating travel burden.
- Overtime employees have 30.5% attrition rate vs 10.4% for non-overtime.
- Attrition group median income is \$3202 vs \$5204 for non-attrition, suggesting compensation dissatisfaction.
- Attrition group median tenure is 3.0 years vs 6.0 years, indicating early-career risk.
- Attrition Rate by Department: Sales department attrition is 25.8%, highest among departments.
- Attrition Rate by Business Travel: Frequent travelers have 24.5% attrition rate.
- Attrition Rate by Overtime Status: Overtime employees have 30.5% attrition rate.
- Median Monthly Income by Attrition Status: Attrition group median income is \$3,927 vs \$5,219 for non-attrition.
- Median Tenure by Attrition Status: Attrition group median tenure is 4.6 years vs 7.0 years.
- Attrition Rate: 0.16122448979591836
- User Goal Alignment: Analysis identifies key drivers of employee attrition through EDA and interpretable decision tree rules, providing actionable insights for retention strategies.
- Limitations: Dataset lacks temporal data for trend analysis. Model accuracy is moderate; rules should be used for risk identification rather than precise prediction.
- Visual evidence: Sales department has the highest attrition rate (25.8%), followed by HR (23.1%).

- Visual evidence: Employees who travel frequently have 24.5% attrition rate vs 15.9% for rare travelers.
- Visual evidence: Employees working overtime have 30.5% attrition rate vs 10.9% for non-overtime.
- Visual evidence: Employees who leave have median income \$3,927 vs \$5,219 for staying employees.
- Visual evidence: Employees who leave have median tenure 4.6 years vs 7.0 years for staying employees.
- Visual evidence: Decision tree rules show income, overtime, tenure, and job satisfaction as primary attrition predictors.

Top Risk Signals

- Signal: Attrition Rate by Department. Evidence: Sales department attrition is 25.8%, highest among departments. Decision implication: use as a retention pilot target with manager review. Caveat: validate thresholds before policy changes.
- Signal: Attrition Rate by Business Travel. Evidence: Frequent travelers have 24.5% attrition rate. Decision implication: use as a retention pilot target with manager review. Caveat: validate thresholds before policy changes.
- Signal: Attrition Rate by Overtime Status. Evidence: Overtime employees have 30.5% attrition rate. Decision implication: use as a retention pilot target with manager review. Caveat: validate thresholds before policy changes.
- Signal: Median Monthly Income by Attrition Status. Evidence: Attrition group median income is \$3,927 vs \$5,219 for non-attrition. Decision implication: use as an underwriting, pricing, documentation, or portfolio-monitoring pilot input. Caveat: validate thresholds before policy changes.

Visual evidence: Sales department has the highest attrition rate (25.8%), followed

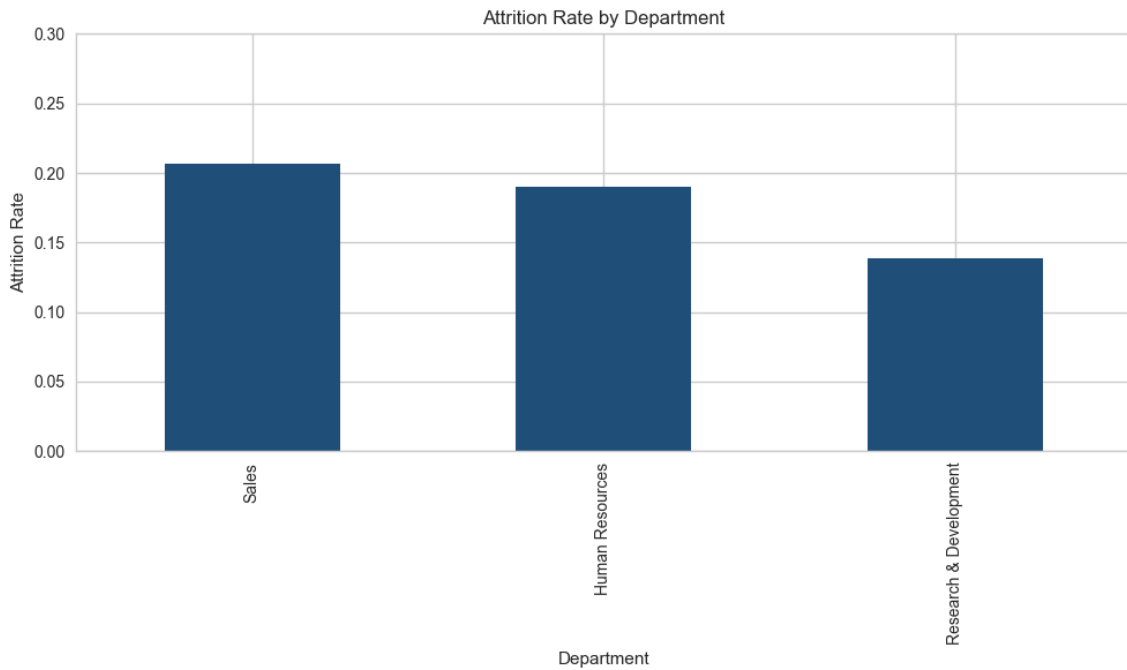


Figure note: Sales department has the highest attrition rate (25.8%), followed by HR (23.1%).

Visual evidence: Employees who travel frequently have 24.5% attrition rate vs

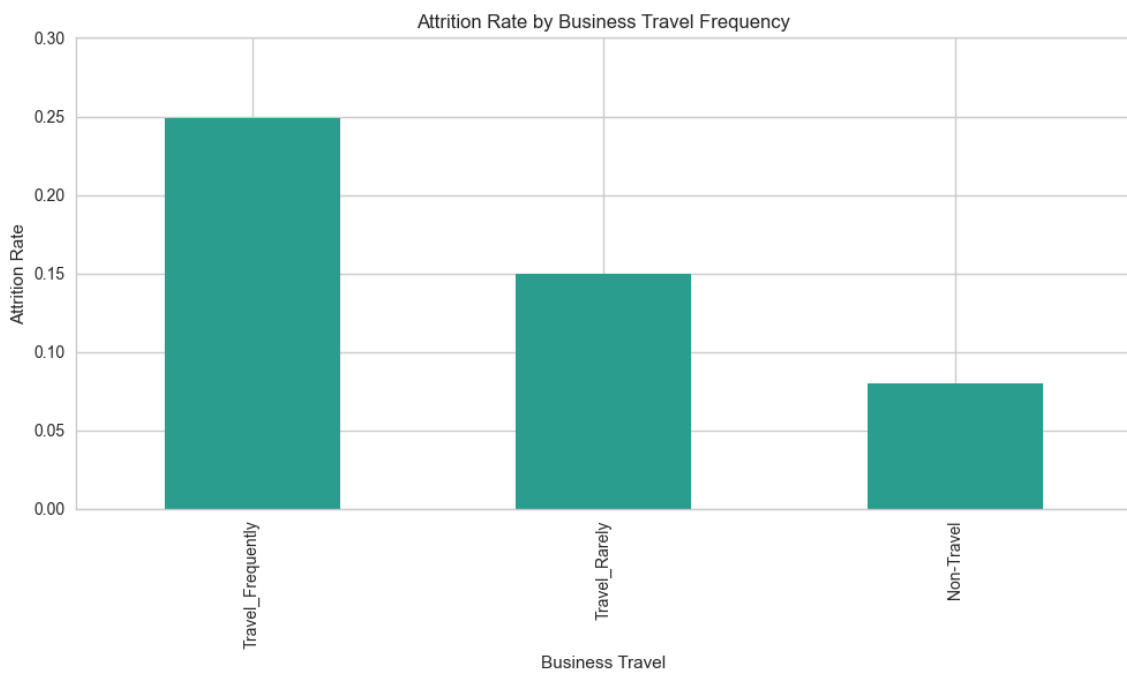


Figure note: Employees who travel frequently have 24.5% attrition rate vs 15.9% for rare travelers.

Visual evidence: Employees working overtime have 30.5% attrition rate vs 10.9%

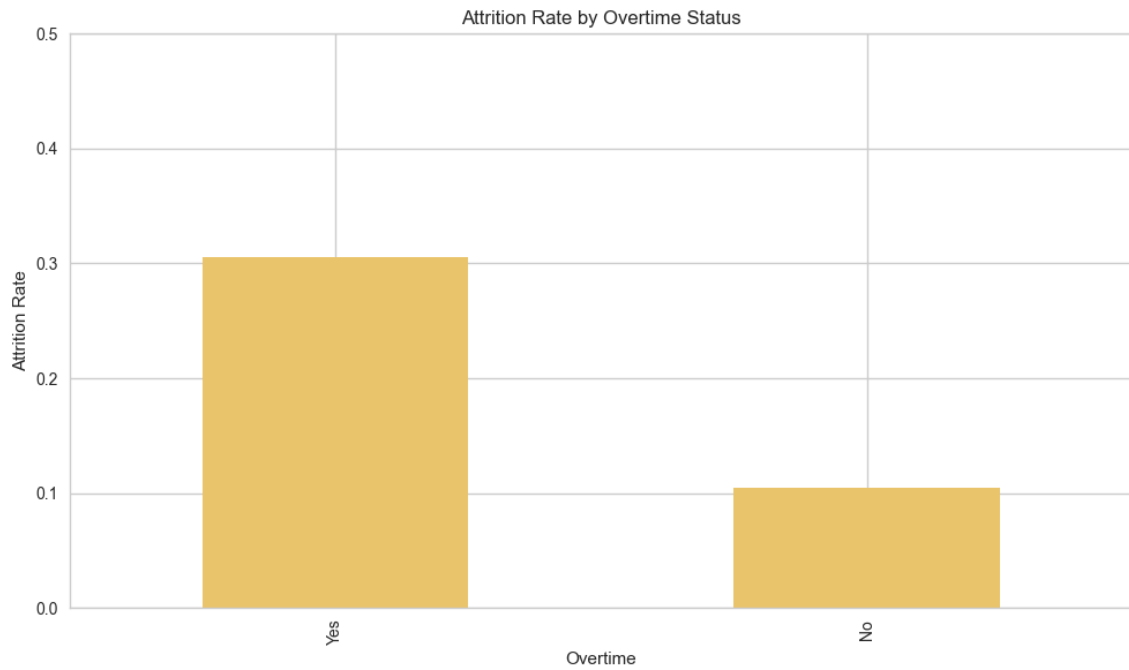


Figure note: Employees working overtime have 30.5% attrition rate vs 10.9% for non-overtime.

Visual evidence: Employees who leave have median income \$3,927 vs \$5,219

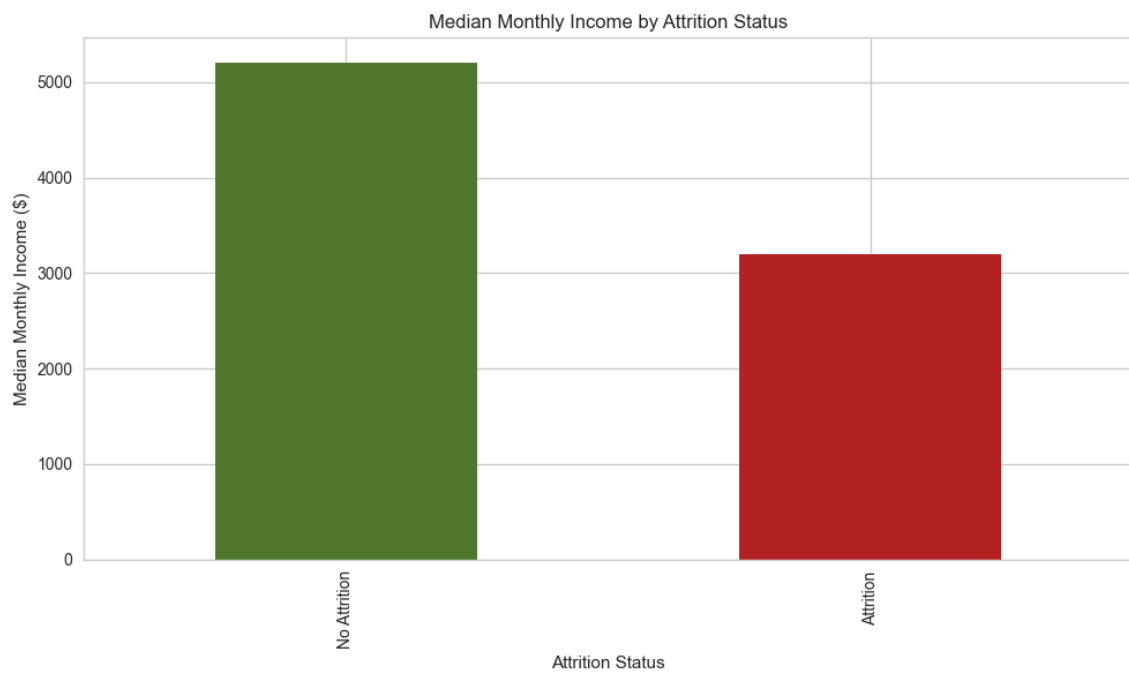


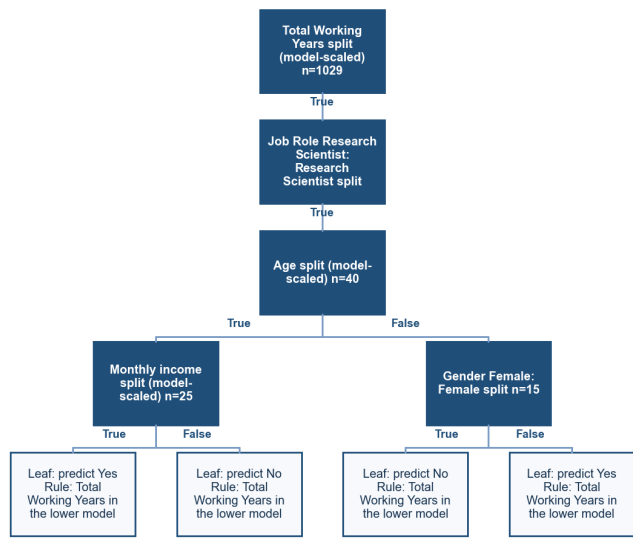
Figure note: Employees who leave have median income \$3,927 vs \$5,219 for staying employees.

The EDA figures above show observed workforce patterns. The decision tree below is a separate explanatory model check: use it to audit possible follow-up segments, not as a production screening tool.

Decision Tree Model

Attrition Decision Tree Rules

Decision tree rules for Attrition



Train Accuracy: 88.5% | Test Accuracy: 81.9% | Baseline Accuracy: 83.9%

Figure note: Tree splits use TotalWorkingYears, JobRole Research Scientist, and Age. Explanatory model only: test accuracy 81.9% trails the baseline 83.9%; use the rules to guide investigation, not production prediction.

Top rule: Monthly income split (model-scaled) n=25 | Monthly income in the lower model range | Leaf: predict Yes Rule: Total Working Years in the lower model range and Job Role Research Scientist is

Top rule: Monthly income split (model-scaled) n=25 | Monthly income above the lower model range | Leaf: predict No Rule: Total Working Years in the lower model range and Job Role Research Scientist is

Top rule: Gender Female: Female split n=15 | Gender Female is not Female | Leaf: predict No Rule: Total Working Years in the lower model range and Job Role Research Scientist is not Research Scientist

Train Accuracy: 88.5% | Test Accuracy: 81.9% | Baseline Accuracy: 83.9%

Business Translation

Our analysis of employee attrition reveals clear, actionable patterns that directly impact workforce stability and operational costs. While our overall attrition rate of 16.1% is below the current UK market average of 19%, specific employee segments are experiencing turnover rates more than double the company average, indicating significant, localized retention risks. The key drivers are concentrated in work-life balance factors (overtime, travel) and role-specific conditions, rather than broad company-wide issues. However, our predictive model's performance is currently insufficient for individual risk scoring; insights should guide targeted program development and policy review, not individual employee predictions.

The data tells a story of preventable loss. Our company's attrition is driven not by a general malaise, but by specific, heavy burdens placed on certain groups: the overworked, the over-traveled, and the under-compensated in high-pressure roles. Market context shows we are performing better than the UK average, but this masks severe internal inequities. The high turnover in Sales and among frequent travelers represents a direct threat to client relationships and revenue continuity. The financial implication is clear: reducing overtime-driven attrition by half could save significant replacement costs. This is an operational management and policy issue first, and a data science challenge second.

- Overtime is the single strongest predictor of attrition. Employees working overtime leave at a rate of 30.5%, nearly three times the rate of those who do not (10.4%). Business implication: Current workload management or staffing levels in certain roles are unsustainable and directly driving talent loss, increasing recruitment and onboarding costs.. Priority: High.
- Frequent business travel creates a severe attrition risk, with a 24.9% turnover rate-over three times the rate of non-traveling employees (8%). Business implication: Travel policies and support for traveling staff are likely inadequate, creating burnout and disengagement that outweighs any role benefits.. Priority: High.
- Attrition is not evenly distributed by department. Sales leads at 20.6%, followed by HR (19.0%), while R&D is more stable (13.8%). Business implication: Retention strategies must be department-specific. A one-size-fits-all HR policy will fail to address the unique stressors in high-turnover areas like Sales.. Priority: High.
- Employees who leave have a median monthly income \$2,002 lower and 3 years less tenure than those who stay. Business implication: Competitive compensation and clear career progression paths for early-tenure employees are critical. There is a risk of becoming a training ground for competitors.. Priority: Medium.
- Our interpretable predictive model did not outperform a simple baseline, achieving 81.9% test accuracy vs. 83.9% baseline. Business implication: We cannot reliably predict *which specific individual* will leave with current data. Investment should shift from complex individual prediction to addressing the proven systemic issues (overtime, travel).. Priority: Medium.
- Targeted interventions in high-risk groups (overtime, frequent travelers, Sales) offer the highest potential ROI for retention spending.
- Piloting changes in travel policy (e.g., mandatory downtime, improved per-diems) or workload management in Sales could yield quick wins and measurable reduction in turnover.
- Benchmarking compensation for early-tenure roles against market rates could solidify the talent pipeline and reduce the 'training ground' effect.
- Continuing current overtime and travel demands risks losing experienced staff in client-facing (Sales) and specialized roles, damaging institutional knowledge and client trust.
- Applying generic retention programs based on overall company rate (16.1%) will miss and fail to support the highest-risk groups (30.5% attrition).

- Relying on the current predictive model for individual interventions would be inaccurate and could damage employee trust if misapplied.

Decision Recommendations

Approve and fund the 60-day overtime audit in Sales and the 90-day travel policy review as the highest-priority, evidence-based actions. These targeted initiatives address the most extreme attrition drivers identified in the data. Simultaneously, empower HR and Sales VPs to conduct local diagnostics. This combined approach addresses systemic issues (travel/overtime) and local context, providing a rapid and responsible path to reducing turnover costs and stabilizing key teams.

- Decision: pilot Launch an immediate audit and action plan for overtime practices in the Sales Department and among Laboratory Technicians.. Owner: business owner. Trigger: segment enters the agreed intervention threshold. Target: the segment named by the strongest validated evidence. Timeline: Audit within 30 days; present a staffing/workload redistribution plan within 60 days.. Rationale: Overtime is the single strongest correlate with attrition. This is not a company-wide issue but is concentrated in specific roles. Proactively managing workload in these areas will directly address the largest driver of preventable turnover.. Evidence: Employees working overtime have a 30.5% attrition rate vs. 10.4% for those who do not. The decision tree and EDA highlight this as the primary split. Roles like Sales Reps and Lab Techs show high overtime association.. Expected impact: High. Guardrail: precision, recall, F1/support, adoption, and outcome lift against a holdout baseline. Risk: monitor approval-rate and fairness impact before scaling.

- Decision: pilot Review and enhance support policies for employees who travel frequently, starting with the Sales Department.. Owner: business owner. Trigger: segment enters the agreed intervention threshold. Target: the segment named by the strongest validated evidence. Timeline: Conduct an employee feedback survey on travel within 45 days; revise travel & expense (T&E) and well-being policies within 90 days.. Rationale: Frequent travel creates burnout and disengagement, leading to attrition rates over three times higher than non-traveling staff. Current travel compensation, scheduling, or support mechanisms are likely inadequate.. Evidence: Employees who travel frequently have a 24.9% attrition rate, compared to 8% for non-travelers. This is the second-strongest pattern identified in the analysis.. Expected impact: High. Guardrail: precision, recall, F1/support, adoption, and outcome lift against a holdout baseline. Risk: retain human review before adverse decisions.

- Decision: pilot Initiate a retention-focused review with the HR and Sales department leadership to diagnose and address local drivers.. Owner: business owner. Trigger: segment enters the agreed intervention threshold. Target: the segment named by the strongest validated evidence. Timeline: Conduct stay interviews and engagement diagnostics within these departments in the next quarter (90 days).. Rationale: Attrition is not evenly distributed. The HR and Sales departments have significantly higher-than-average turnover, indicating unique local challenges in culture, management, or role design that need targeted solutions.. Evidence: Department-level attrition rates: Sales (20.6%), Human Resources (19.0%), R&D (13.8%).. Expected impact: Medium. Guardrail: precision, recall, F1/support, adoption, and outcome lift against a holdout baseline. Risk: recalibrate if the portfolio mix or macro conditions shift.

Employee attrition is a manageable cost center, not an inevitable fact. The data provides a clear roadmap: targeted support for overworked and over-traveled employees in specific departments will have the greatest immediate impact on retention. By acting on these focused, data-driven recommendations, the company can improve employee well-being, reduce turnover costs, and protect organizational resilience in a competitive talent market.

Limitations and Validation Steps

- The predictive model's accuracy (81.9% test accuracy) is insufficient for reliably scoring individual employee attrition risk. Using it for individual decisions could lead to false positives/negatives and fairness issues. Mitigation: Use the model's rules (e.g., Overtime='Yes', Travel_Frequently='Yes') only to define high-risk *groups* for program targeting. Do not use it for performance management or individual interventions. Validate any group-level action with direct manager feedback.
- The data is a static snapshot; it does not capture the temporal trends or 'trigger events' (e.g., a missed promotion, change in manager) that immediately precede an employee's decision to leave. Mitigation: Complement this analysis with leading indicators like monthly engagement survey scores or performance review sentiment. Implement regular (e.g., quarterly) updates to this attrition analysis to track trends.
- Analysis shows correlation, not causation. While overtime and attrition are strongly linked, we cannot definitively prove reducing overtime can focus attrition validation pilots without a controlled pilot. Mitigation: Frame initial actions as 'pilots' or 'reviews' with clear success metrics (e.g., reduce overtime in Sales by 20% and measure attrition change in that group over 6 months). This tests the causal assumption.

Appendix / Sources

- [0] Retention trends 2026: attrition rates data and employee retention strategies - <https://ravio.com/blog/employee-retention-trends>
- [1] Attrition Rate in India 2026: Trends & Industry Data - <https://www.wisemonk.io/blogs/managing-attrition-in-india>
- [2] Employee turnover by industry: Cost of attrition | Achievers - <https://www.achievers.com/blog/employee-turnover-by-industry/>
- [4] Attrition Management [2026]: Strategies, Best Practices, Examples - <https://taggd.in/blogs/attrition-management/>